Forecasting Monthly Foreign Tourist Arrivals and Seasonal Clustering of Source Countries in India

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Abstract

This study investigates foreign tourist arrivals (FTAs) in India through two related tasks: forecasting national monthly arrivals and identifying source-country groups with similar seasonal distributions. The monthly dataset contains 72 observations from January 2019 to December 2024. A chronological split uses 2019–2023 for training and 2024 for testing. Among the evaluated forecasting approaches, the Seasonal Naive benchmark achieved the lowest reported holdout error (MAE = 43,253.83; RMSE = 58,173.06; R^2 = 0.85; MAPE = 4.89%). Linear Regression was the strongest of the four assigned regression models (MAE = 51,318.60; RMSE = 59,543.55; R^2 = 0.84; MAPE = 5.96%). Country segmentation was performed using four quarterly percentage shares for 183 country/territory profiles with at least 100 annual arrivals. K-Means with K = 3 achieved a silhouette score of 0.3387 and identified Q1/Q4-weighted, comparatively balanced and Q3-weighted profiles. A complete monthly-by-country matrix was unavailable in the selected official source; quarterly shares are therefore used as an explicit proxy for country-level seasonality. The results are exploratory in view of the short time series, COVID-19 disruption and moderate-to-weak cluster separation.

Keywords—Foreign Tourist Arrivals, India, Time-Series Forecasting, Linear Regression, Seasonal Naive, SARIMA, Holt-Winters, K-Means, Quarterly Seasonality.

1. Introduction

Tourism demand changes over time because of seasonal travel preferences, public holidays, economic conditions, air connectivity, visa policies and unexpected disruptions. Reliable forecasts can support capacity planning for transport, accommodation, staffing and destination management. Country-level seasonal profiles can further help tourism planners understand whether source markets concentrate visits in different parts of the year.

India's 2019–2024 monthly series contains the exceptional COVID-19 shock and the subsequent recovery. This paper examines whether classical machine-learning and time-series methods can capture national monthly arrivals and whether source countries can be grouped using their seasonal distribution. The work contributes (i) a chronological forecasting evaluation, (ii) a comparison of assigned regression models with seasonal and statistical benchmarks, and (iii) a country-wise seasonal segmentation using quarterly shares where full monthly country-wise observations are unavailable.

1.1 Problem Definition and Objectives

The problem is to forecast monthly foreign tourist arrivals in India and identify groups of source countries with similar seasonal arrival patterns. The objectives are to describe trend and seasonality, quantify pandemic-era decline and recovery, compare forecasting models on an unseen year, and interpret quarterly country clusters.

2. Related Work

Seasonal-naive forecasting is a strong baseline when annual seasonal patterns persist [2]. The ARIMA family models temporal dependence through autoregressive, moving-average and differencing terms; seasonal ARIMA extends this structure to repeating seasonal periods [3]. Holt-Winters exponential smoothing represents level, trend and seasonality. K-Means partitions observations by minimising within-cluster squared distances [4], while the silhouette coefficient evaluates cohesion and separation [5]. These interpretable methods are appropriate for an undergraduate study because their assumptions and outputs can be examined directly.

3. Dataset and Preprocessing

3.1 National Monthly FTA Series

The national dataset contains 72 monthly observations from January 2019 through December 2024. The response variable is monthly foreign tourist arrivals. Dates were converted to a datetime representation, sorted chronologically and checked for missing target values. No missing target observations were found. The first 60 months (January 2019–December 2023) form the training set; the final 12 months (January–December 2024) form the holdout test set.

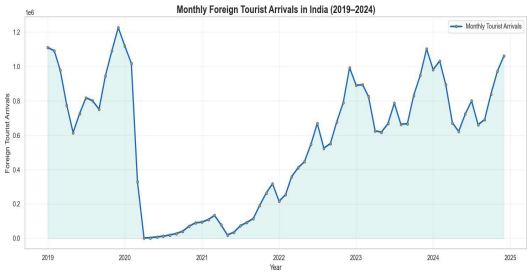


Fig. 1. Monthly foreign tourist arrivals in India, 2019–2024 (executed notebook output).

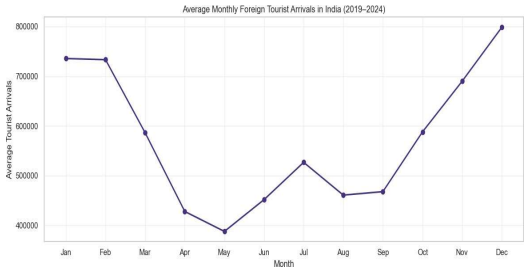


Fig. 2. Average monthly seasonal pattern of foreign tourist arrivals.

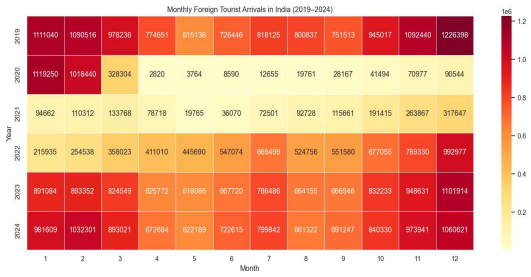


Fig. 3. Year–month heatmap showing the seasonal pattern and pandemic-era disruption.

3.2 Country-wise Quarterly Dataset

The source-country data contain 209 country/territory profiles with annual arrival totals and four quarterly percentage shares: Q1 (January–March), Q2 (April–June), Q3 (July–September) and Q4 (October–December). Applying a minimum annual-arrival threshold of 100 retains 183 profiles and excludes 26 low-volume profiles.

Crucially, annual arrival totals are used only for the low-volume filtering rule. They are not clustering features. K-Means receives only the four quarterly percentage shares, so the distance measure reflects seasonal shape rather than the absolute size of a source market. The selected official publication does not provide a complete monthly-by-country matrix for all markets. The quarterly profiles are therefore a transparent proxy and must not be described as monthly country-wise clustering.

3.3 Preprocessing and Seasonal Diagnostics

The forecasting split is chronological and is not shuffled. The Augmented Dickey–Fuller (ADF) test on the original training series produced statistic –2.1302 and p-value 0.2325; the unit-root null was not rejected at the 5% level. After first-order differencing, the statistic was –4.7237 with p-value 0.0000759, supporting stationarity of the differenced series. Notebook plots document the monthly pattern, COVID-19 disruption, training/test split, differencing and autocorrelation diagnostics.

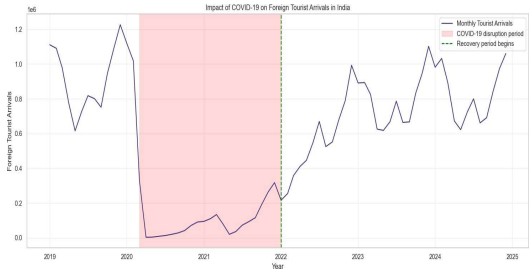


Fig. 4. Monthly arrivals with the COVID-19 disruption period highlighted.

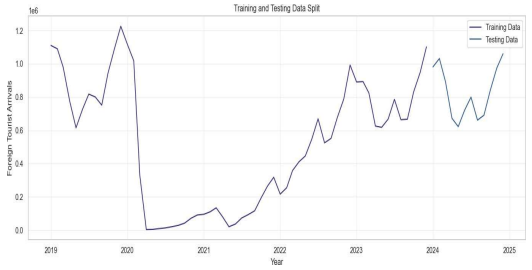


Fig. 5. Chronological training and 2024 testing split.

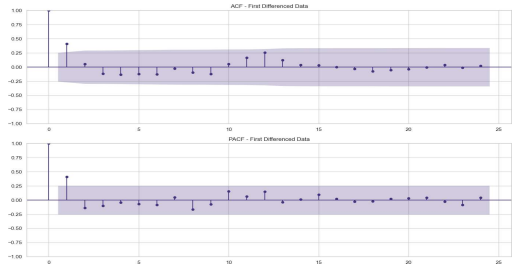


Fig. 6. First-order differencing used in the stationarity analysis.

4. Methodology

4.1 Forecasting Models

Four assigned regression approaches were evaluated: Linear Regression, degree-2 Polynomial Regression, Ridge Regression and k-Nearest Neighbours (k-NN) Regression. The supplementary implementation constructs time-related predictors, COVID/recovery indicators and monthly seasonal dummy variables. Hyperparameter selection for Ridge and k-NN uses expanding-window TimeSeriesSplit on the training period only. The Seasonal Naive benchmark predicts each month in 2024 using the corresponding month in 2023. Additional benchmarks include SARIMA candidates with a 12-month seasonal period and additive/multiplicative Holt-Winters smoothing.

4.2 Evaluation Measures

Mean Absolute Error (MAE) is the average absolute difference between actual and predicted arrivals. Root Mean Squared Error (RMSE) penalises large forecast misses more strongly. Mean Absolute Percentage Error (MAPE) expresses absolute error relative to actual arrivals as a percentage. R^2 measures the proportion of variation explained relative to the test-set mean. Lower MAE, RMSE and MAPE are preferred; higher R^2 is generally preferred.

4.3 K-Means Seasonal Clustering

The clustering matrix consists of four percentage-share variables, Q1–Q4. Since all features are percentages on the same scale, percentage-point distances are used directly. Candidate values of K are compared using inertia and silhouette scores. K = 3 is selected because its silhouette score (0.3387) is slightly higher than K = 2 (0.3175). Cluster centroids are interpreted as average quarterly shares.

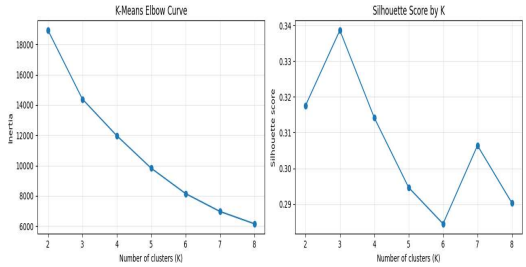


Fig. 10. Elbow and silhouette analysis for selecting the number of clusters.

5. Experimental Setup

The work was implemented in Python using Jupyter Notebook, pandas and NumPy for data preparation, Matplotlib/Seaborn for visualisation, scikit-learn for regression and clustering, and statsmodels for time-series diagnostics and statistical forecasting. The original notebook contains exploratory analysis, ADF diagnostics, Seasonal Naive, SARIMA and Holt-Winters experiments. A supplementary notebook contains the assigned regression comparison and corrected quarterly-only K-Means implementation. The test period is fixed to calendar year 2024 to simulate forecasting future observations.

6. Results

6.1 Pandemic Impact and Recovery

Table 1. Annual average monthly foreign tourist arrivals and change relative to 2019.

Year	Mean monthly FTAs	Change from 2019 (%)
2,019.00	910,862.92	0.00
2,020.00	228,730.50	-74.89
2,021.00	127,259.50	-86.03
2,022.00	536,455.58	-41.10
2,023.00	793,410.67	-12.89
2,024.00	829,310.17	-8.95

Average monthly arrivals declined by 74.89% in 2020 and 86.03% in 2021 relative to 2019. The series recovered from 2022 onward. The 2024 average monthly level was approximately 91.05% of the 2019 baseline, indicating substantial but incomplete recovery within the study period.

6.2 Forecasting Performance

Table 2. Assigned regression model performance on the 2024 holdout period.

Model	MAE	RMSE	R2	MAPE (%)
Seasonal Naive baseline	43,253.83	58,173.06	0.85	4.89
Linear Regression	51,318.60	59,543.55	0.84	5.96
Ridge Regression	217,326.72	223,503.46	-1.22	26.02
k-NN Regression	297,675.96	303,924.32	-3.11	35.82
Polynomial Regression (degree 2)	393,637.47	402,087.60	-6.19	48.69

The Seasonal Naive benchmark achieved the lowest overall reported error (MAE = 43,253.83; RMSE = 58,173.06; R^2 = 0.85; MAPE = 4.89%). Linear Regression was the strongest of the four assigned regression approaches (MAE = 51,318.60; RMSE = 59,543.55; R^2 = 0.84; MAPE = 5.96%). This result shows why a simple seasonal baseline must be included: repeating the previous year's corresponding month can outperform more complex methods when the annual pattern persists.

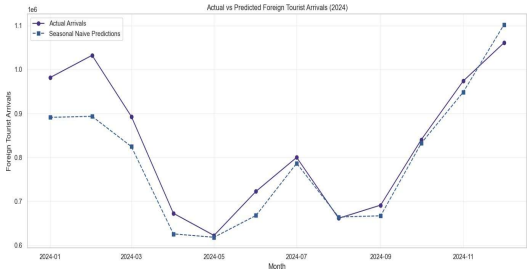


Fig. 8. Forecast comparison for the 2024 holdout period.

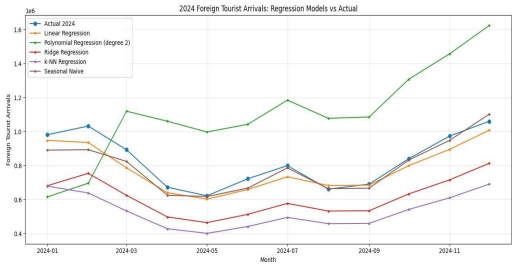


Fig. 9. Actual and predicted 2024 arrivals for the assigned regression models.

Table 3. Additional time-series model results on the 2024 holdout period.

Model	MAE	RMSE	MAPE (%)
SARIMA(1,1,0)(0,0,1,12)	180,538.48	202,320.41	24.47

Model	MAE	RMSE	MAPE (%)
SARIMA(1,1,0)(0,1,1,12)	55,328.15	63,884.34	6.92
SARIMA(0,1,1)(0,1,1,12)	62,652.37	71,218.70	7.89
SARIMA(1,1,1)(0,1,1,12)	63,220.49	71,778.36	7.96
Holt-Winters additive	146,856.41	165,630.58	18.86
Holt-Winters multiplicative	294,201.47	327,963.73	37.06

Seasonal differencing improved the evaluated SARIMA configuration: SARIMA(1,1,0)(0,1,1,12) achieved MAPE 6.92%, compared with 24.47% for the candidate without seasonal differencing. The Seasonal Naive benchmark nevertheless remained the best overall model. Additive and multiplicative Holt-Winters variants produced larger errors.

6.3 Country-wise Seasonal Clusters

Table 4. K-Means cluster sizes and mean quarterly shares for K = 3.

Cluster	Country_Count	Q1_percent	Q2_percent	Q3_percent	Q4_percent
1.00	61.00	33.49	16.05	18.01	32.45
2.00	108.00	26.24	21.04	25.40	27.32
3.00	14.00	15.76	14.37	41.90	27.96

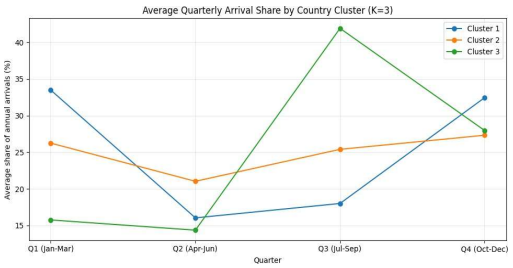


Fig. 11. Mean quarterly arrival-share profiles for the three country clusters.

Cluster 1 contains 61 profiles and is relatively Q1/Q4 weighted (33.49% and 32.45%). Cluster 2 contains 108 profiles and is comparatively balanced across the four quarters. Cluster 3 contains 14 profiles and is strongly Q3 weighted, with approximately 41.90% of annual arrivals in July–September. The K = 3 silhouette score of 0.3387 indicates moderate-to-weak separation; the clusters are broad exploratory segments rather than sharply distinct groups.

7. Discussion

At the national level, strong monthly variation suggests that tourism capacity and promotion should be planned seasonally rather than assuming uniform demand. The Seasonal Naive result is a useful operational reference: before deploying a more complex model, planners should test whether it improves on a simple previous-year seasonal forecast. The quarterly country profiles suggest that source-market campaigns may be timed differently for Q1/Q4-weighted and Q3-weighted segments. These are planning hypotheses based on observed distributions, not causal findings.

7.1 Limitations

(i) The monthly national series contains only 72 observations, with 60 training and 12 test points. (ii) COVID-19 introduces a structural break that may reduce the stability of historical relationships. (iii) Country-wise clustering is quarterly because a complete monthly-by-country matrix was unavailable. (iv) Twenty-six low-volume profiles were excluded using the 100-arrival threshold. (v) The K = 3 silhouette score indicates limited cluster separation. (vi) External predictors such as exchange rates, flight connectivity, visa policies, holidays and major events were not incorporated.

7.2 Future Work

Future work should extend the monthly series, evaluate models with multiple rolling-origin test windows, obtain monthly country-level data if officially released, test the sensitivity of clusters to the 100-arrival threshold, compare hierarchical and density-based clustering, and include relevant external variables.

8. Conclusion

This study analysed India's foreign tourist arrivals using national monthly forecasting and source-country seasonal segmentation. Arrivals declined sharply during the pandemic and recovered to approximately 91.05% of the 2019 average monthly level by 2024. Seasonal Naive was the best overall 2024 forecasting method, while Linear Regression was the strongest assigned regression model. Quarterly-only K-Means identified three broad seasonal profiles among 183 retained source markets. The study demonstrates the importance of chronological evaluation, baseline comparison and transparent acknowledgement of data granularity.

Declarations

Data availability. Prepared monthly FTA and country-wise quarterly-share datasets, notebooks and results workbook are included with this submission. The source is the Ministry of Tourism, Government of India, India Tourism Data Compendium 2025, including the relevant month-wise, peak/lean-month and quarterly country-distribution tables.

Conflict of interest. The author declares no conflict of interest. Funding. No external funding was received. Use of AI tools. AI assistance supported code explanation, debugging and report drafting. The student should rerun all notebooks, verify the reported values and follow institutional disclosure requirements.

References

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Appendix A. Reproducibility

The analysis can be repeated by running the two supplied Jupyter notebooks with the accompanying CSV files. The first notebook covers exploratory analysis and time-series models, while the second covers the assigned regression models and quarterly-share K-Means clustering.

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